COMP438/538 Reinforcement Learning

Model-Free Prediction

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Models and Prediction without the Models

So far:

- MDPs
  - State, Action, Reward, Transition Dynamics, Discount
  - Optionally terminal and start states
- Returns, Values and Q-Values

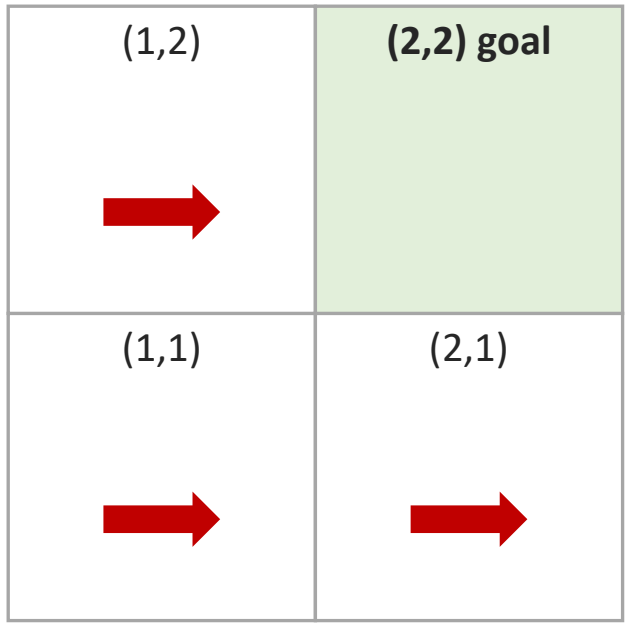
- Policy Evaluation
- Policy Iteration (Evaluation then Selection)
- Value Iteration (Concurrent Evaluation and Selection)

- Which computations need the transition and reward model?
- Which action-selection step can use a learned table directly?

Policy evaluation without the model

- With a model: evaluate a policy, then improve it.
- Today: evaluate the **same fixed policy** using observed experience.
- Next week: improve policies using experience.

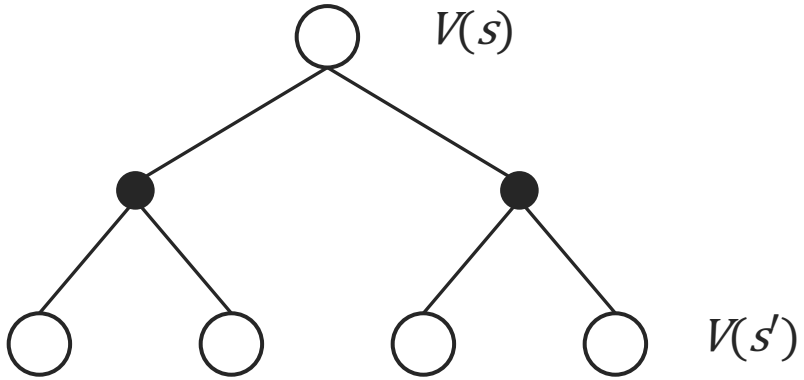
Question: You can run this policy and observe what happens. How could you estimate $v^{\pi}((1,1))$?

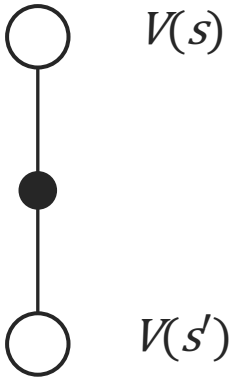


Always choose right; γ=0.5.

Transition probabilities: **?**

Why learn from samples?





Expected backup	Sample backup
Average over possible outcomes using the model	Use an observed transition
Need outcome probabilities and expected rewards	Receive $(S_t, A_t, R_{t+1}, S_{t+1})$

- A robot's dynamics may be unknown or unreliable.
- A simulator may produce samples even when enumerating all outcomes is impractical.
- One cheap update can still require many samples to learn accurately.

Reinforcement Learning: Models are unknown ahead of time (Today: Sampling, Later: Model-Learning)

Side Note: Three different questions about the data

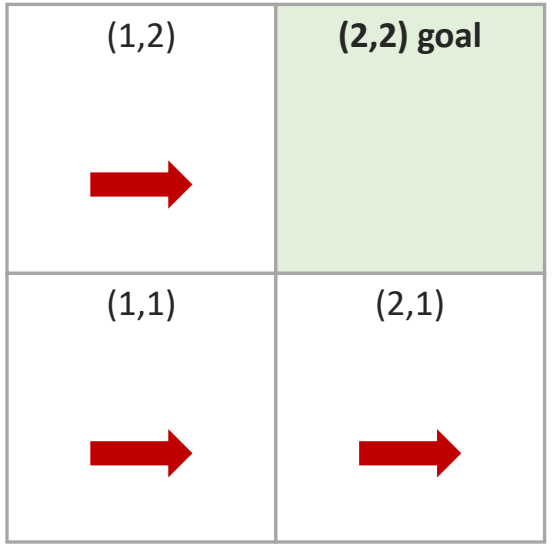
Question	Distinction
Do we use or learn an environment model?	Model-based / model-free
Can we collect more interaction data while learning?	Online / offline (fixed data)
Does the behaviour generating data match the policy being evaluated?	On-policy / off-policy

Today: tabular, model-free, on-policy prediction of a fixed π.

One observed episode

(1,1) → (1,1) → (1,2) → goal

t	S_t	A_t	R_{t+1}	S_{t+1}
0	(1,1)	right	−0.04	(1,1)
1	(1,1)	right	−0.04	(1,2)
2	(1,2)	right	+1	goal



Important concepts:

- Transitions
- Trajectories/Roll-outs
- Episode
- Return
- Averaging Returns

Monte Carlo: value as a mean return

$$v^{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s]$$

Monte Carlo prediction: follow π, observe complete episodes, and average returns after visits to each state.

$$G_T = 0, \qquad G_t = R_{t+1} + \gamma G_{t+1}$$

Episode: (1,1) → (1,1) → (1,2) → goal; rewards −0.04, −0.04, +1; $\gamma = 0.5$.

Your turn: calculate G_2, then G_1, then G_0 for our episode.

$$G_2 = 1, \qquad G_1 = -0.04 + 0.5(1) = 0.46, \qquad G_0 = -0.04 + 0.5(0.46) = 0.19$$

One observed return is a sample, not the expectation.

Which visits count?

Visit to (1,1)	Return following that visit
First, t=0	0.19
Second, t=1	0.46

Episode: (1,1) → (1,1) → (1,2) → goal

- **First-visit MC:** include only the first visit to a state in each episode.
- **Every-visit MC:** include every visit to that state.

Question: what sample mean does each produce after this one episode? (For (1,1))

Answer: first-visit 0.19; every-visit $(0.19 + 0.46)/2 = 0.325$.

From an average to an incremental update

For one newly included return G_t, first increment $N(s)$:

$$N(s) \leftarrow N(s) + 1$$

$$V(s) \leftarrow \frac{(N(s) - 1)V(s) + G_t}{N(s)} = V(s) + \frac{1}{N(s)}[G_t - V(s)]$$

A common update form

$$V(S_t) \leftarrow V(S_t) + \alpha[U_t - V(S_t)]$$

U_t: target. α: fraction of the discrepancy used for this update.

Step size for MC	Meaning
$1/N(s)^*$	Exact sample mean of included returns
Constant α	Recency weighting; initialization can still matter, better for non-stationary problems

*This is a special case of decayed α that meets the Robbins–Monro conditions for convergence, but not all decay schedules work

First-visit MC prediction

1. Fix π. Initialize $V(s) = 0$ and $N(s) = 0$ for nonterminal states.
2. Generate a complete episode following π.
3. Compute its returns backwards: $G_T = 0$, then $G_t = R_{t+1} + \gamma G_{t+1}$.
4. Scan the episode in time order. At the **first visit** to each state:
   - increment $N(S_t)$;
   - update $V(S_t) \leftarrow V(S_t) + \frac{1}{N(S_t)}[G_t - V(S_t)]$.
5. Repeat with another episode.

For constant-step MC, replace $1/N(S_t)$ by α in step 4. The first-visit rule stays.

Question: How do the steps change if we do the every-visit version?

Driving home: must learning wait?

Predict the minutes remaining until home

Situation	Elapsed minutes	Predicted minutes remaining	Predicted total minutes
Leave office	0	30	30
Reach car; raining	5	35	40
Exit highway	20	15	35
Behind slow truck	30	10	40
Enter home street	40	3	43
Arrive home	43	0	43

At the car, would you revise the office prediction now, or wait until home?

Another Question: What happens if the problem is continuing?

Standalone prediction example: rewards are elapsed minutes; $\gamma = 1$. Book Example 6.1 (Sutton & Barto, 2018).

A sampled Bellman target – Temporal Difference (TD) Learning

1. Recall the Bellman expectation equation:

$$v^{\pi}(s) = \mathbb{E}_{\pi}[R_{t+1} + \gamma v^{\pi}(S_{t+1}) \mid S_t = s]$$

2. Use one observed transition and the current next-state estimate:

$$U_t = R_{t+1} + \gamma V(S_{t+1})$$

3. Compare the target with the current estimate, then update:

$$\delta_t = U_t - V(S_t), \qquad V(S_t) \leftarrow V(S_t) + \alpha\delta_t$$

Bootstrapping: use an existing value estimate in a new update target.

The expected TD target is a Bellman backup of the current V; it need not equal v^{π}.

TD(0) prediction

Fix π and choose α. Initialize the nonterminal estimates; set $V(\text{terminal}) = 0$.

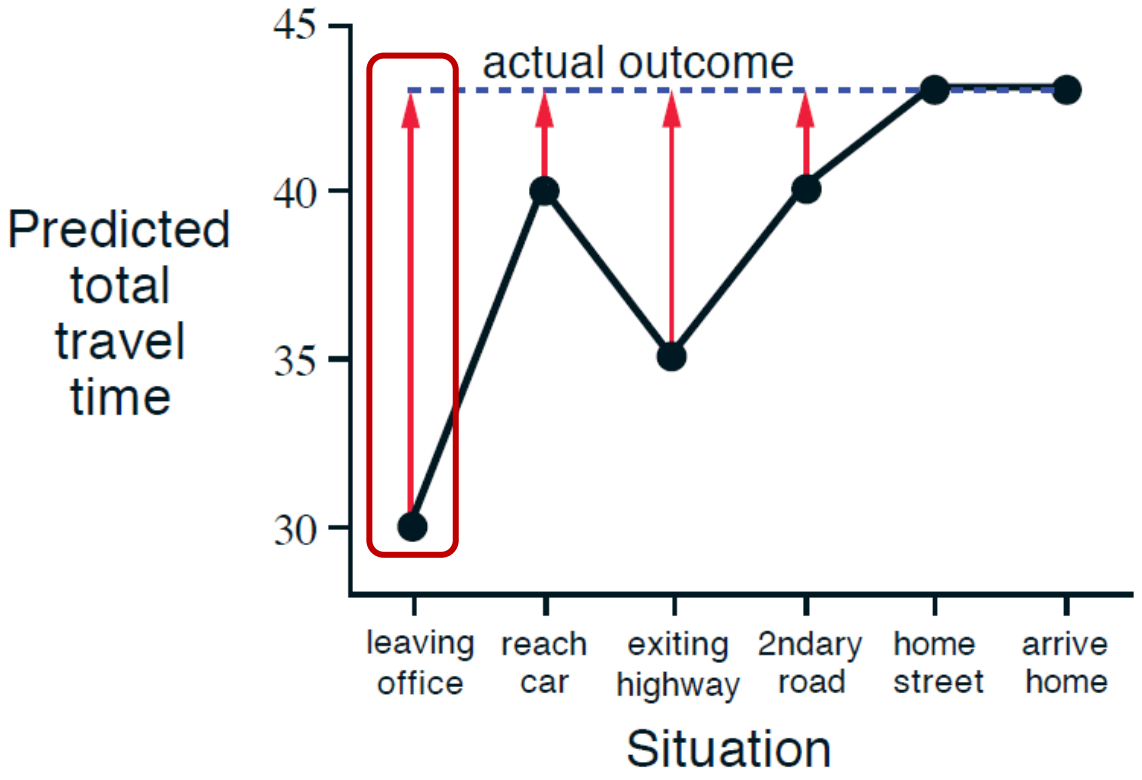
At each transition:

1. Take A_t according to π; observe R_{t+1} and S_{t+1}.
2. Form $U_t = R_{t+1} + \gamma V(S_{t+1})$.
3. Update $V(S_t) \leftarrow V(S_t) + \alpha[U_t - V(S_t)]$.
4. Advance to the next state; if terminal, end the episode.

Update after observing the transition. Read the current table before changing it.

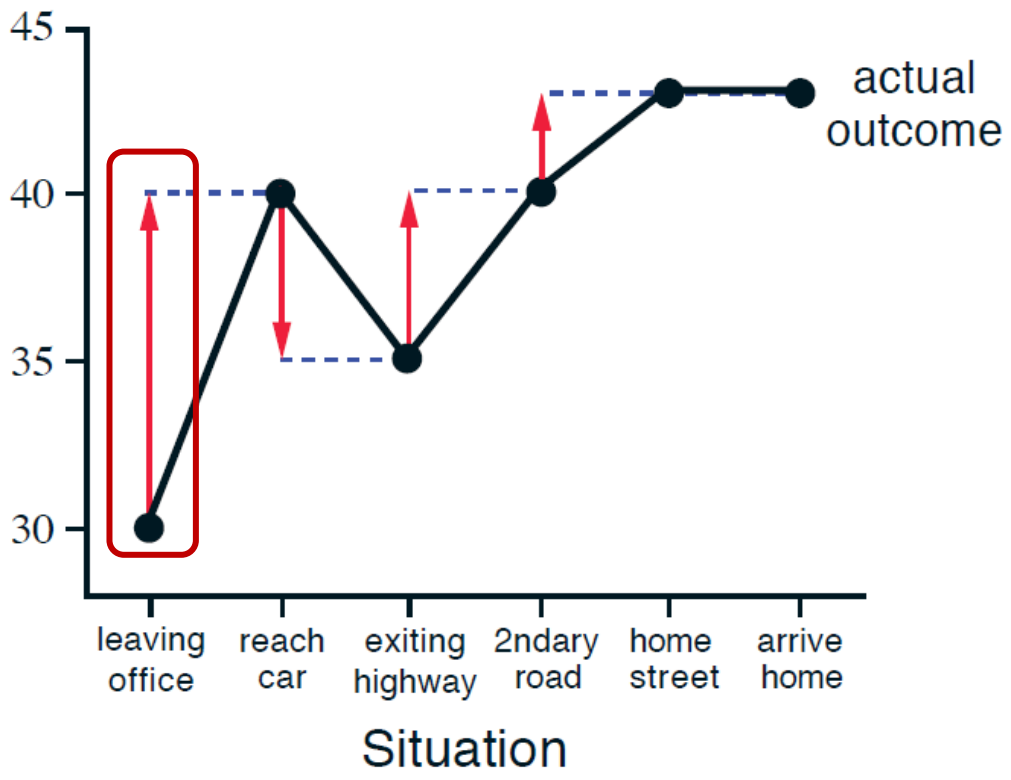
Driving home: what each update changes

MC: toward the final outcome



First arrow: office prediction $30 \to 43$, after reaching home

TD: toward the next prediction



First arrow: office prediction $30 \to 5+35=40$, after reaching the car

$\gamma=1$; $\alpha=1$ for the full arrows. Figure: Sutton & Barto (2018), Figure 6.1.

Your turn: MC and TD on the same episode

(1,1) → (1,1) → (1,2) → goal; rewards −0.04, −0.04, +1; returns 0.19, 0.46, 1

Reset each method independently: $V = 0$, $\gamma = 0.5$, **constant** $\alpha = 0.5$.

1. First-visit MC: update each state's first visit using its completed return.
2. TD(0): update after each transition, in time order.
3. Why does the goal reward affect the estimates differently?

Method / event	Target U_t	Estimate being changed	Value after update
MC, first (1,1) visit	0.19	$V((1,1))$	0.095
MC, first (1,2) visit	1	$V((1,2))$	0.5
TD, t=0	−0.04	$V((1,1))$	−0.02
TD, t=1	−0.04	$V((1,1))$	−0.03
TD, t=2	1	$V((1,2))$	0.5

MC and TD have different targets and update schedules; this is not a performance ranking.

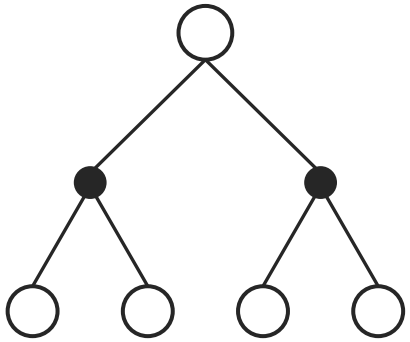
One episode versus an expected value

State	First-visit MC after this episode	TD(0) after this episode	True v^{π} from the known model
(1,1)	0.095	−0.03	≈ 0.051
(1,2)	0.5	0.5	≈ 0.836

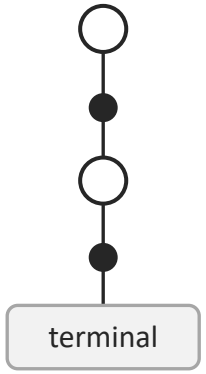
The right-hand column is a checking benchmark; the learner did not use it.

Questions: Can this episode establish which method is better? What would we need to compare learning after many episodes?

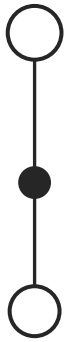
Sampling and bootstrapping



DP: all successors, one step



MC: one path to termination



TD(0): one sampled step

Method	Uses sampled experience?	Bootstraps from value estimates?	What supplies the target?
DP expected evaluation backup	No	Yes	Model-weighted reward and successor values
Complete-return MC	Yes	No	Observed return through termination
TD(0)	Yes	Yes	One observed reward and next-state estimate

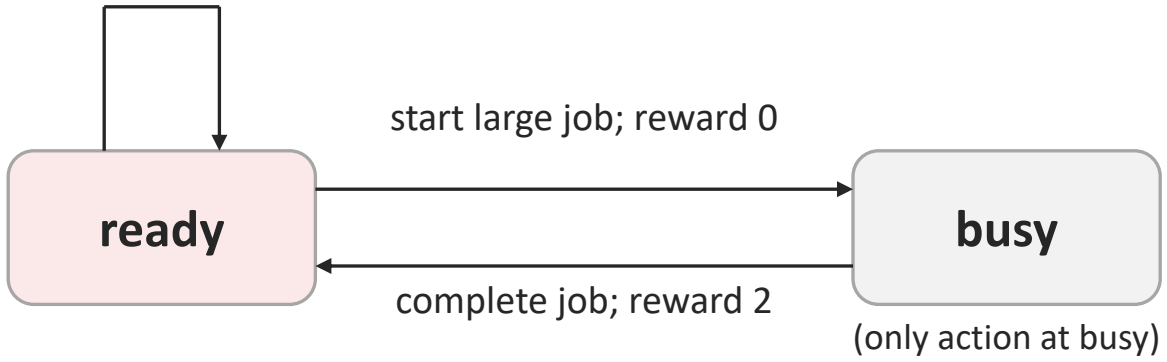
Sampling: use observed outcomes in place of a full expectation. **Bootstrapping:** use a value estimate in the target.

Check: Which needs a model? Which must wait for the completed return? Which can update before the episode ends?

A task that keeps running

A continuing service system — equal-duration time steps

steady service; reward 0.8



Starting from ready:

Fixed policy at ready	Reward sequence
Always steady service	0.8, 0.8, 0.8, …
Always start a large job	0, 2, 0, 2, …

There is no terminal state. What should good long-run performance mean?

Discounted return and average reward

Average reward: expected reward per step over a long run

For the policies in this service example, starting from ready:

$$r(\pi) \doteq \lim_{t\to\infty} \frac{1}{t} \mathbb{E}_\pi[R_1 + R_2 + \cdots + R_t \mid S_0 = \text{ready}]$$

Policy	Average reward per step	Discounted return from ready, γ=0.5
Steady service	0.8	1.6
Large jobs	(0 + 2)/2 = 1	4/3 ≈ 1.333

The objective changes the ranking: discounted return prefers steady service; average reward prefers large jobs.

Self-Study Suggestion: Sutton & Barto §10.3 for the differential return and values that average-reward methods learn.

What can we learn before the episode ends?

- MC updates toward a completed observed return.
- TD updates toward an observed reward plus estimated continuation.
- Both can evaluate a fixed policy from experience.

Which needs the next state's estimate? Which needs the completed return?

Next session: bias, variance, n-step targets and evaluating learning.

Next topic: improve policies using learned action values.

Can one episode decide?

We evaluated a fixed policy from experience:

$$U_t = G_t \quad (\mathrm{MC}), \qquad U_t = R_{t+1} + \gamma V(S_{t+1}) \quad (\mathrm{TD}(0))$$

Our grid episode gave different value updates.

Question: Does one episode tell us which method learns better?

Today: noisy targets, n-step prediction and evidence for a comparison.

Bias and variance describe repeated samples

At the same state, repeat the sampling experiment while holding the current table fixed.

- **Bias:** mean target minus the true value.
- **Variance:** spread of targets around their own mean.
- **Mean squared error = variance + bias².**

Illustration, true value 1.0:

Equally likely targets	Mean	Bias	Variance
0.0 or 2.0	1.0	0.0	1.0
0.8 or 1.0	0.9	−0.1	0.01

What randomness enters each target?

	MC	TD(0)
Target	G_t	$R_{t+1}+\gamma V(S_{t+1})$
Samples	Remaining episode	One transition
Uses the current value table in the target?	No	Yes

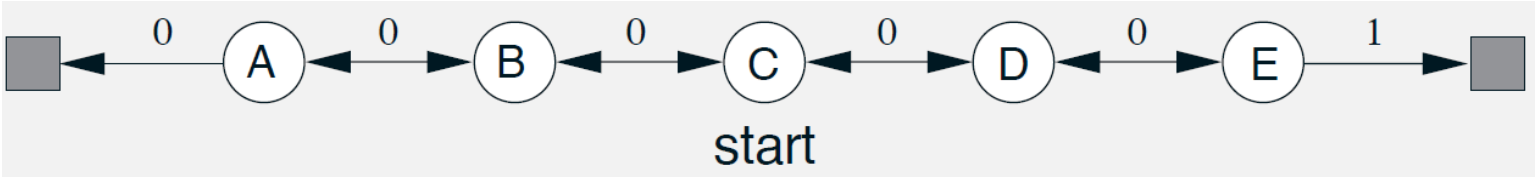
$$\mathbb{E}_\pi[G_t \mid S_t = s] = v^\pi(s)$$

$$\mathbb{E}_\pi[R_{t+1} + \gamma V(S_{t+1}) \mid S_t = s] - v^\pi(s) = \gamma\, \mathbb{E}_\pi[V(S_{t+1}) - v^\pi(S_{t+1}) \mid S_t = s]$$

MC has an unbiased **target**. TD's target can inherit error from the table.

Question: On the grid, which random transitions affect each target for the start state?

Example 6.2: random-walk prediction



Five nonterminal states **A, B, C, D, E** between left and right terminals. Start at **C**. Each step moves left or right with probability 0.5.

- **New example:** $\gamma = 1$, +1 on right-terminal entry, 0 on every other transition.
- Both terminals have zero continuation value.
- Start with $V(s) = 0.5$ for each nonterminal state.

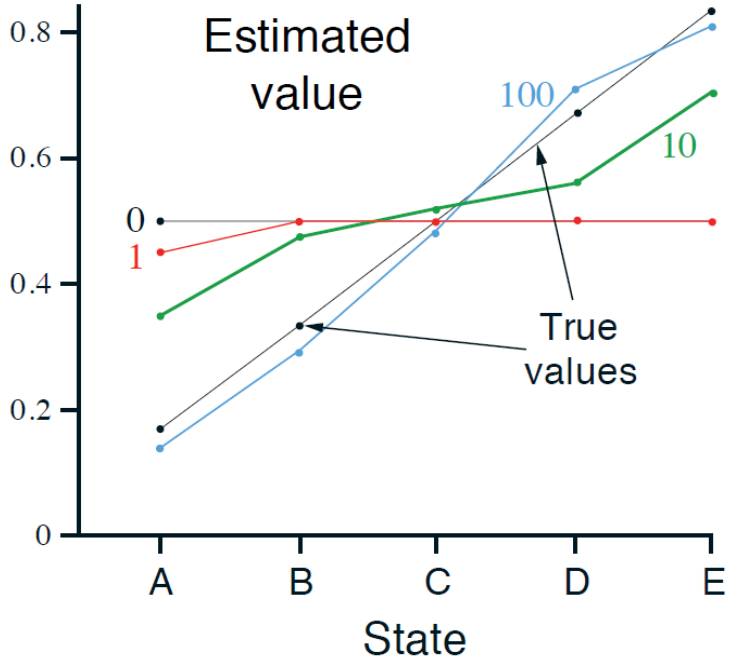
State	A	B	C	D	E
True value	1/6	2/6	3/6	4/6	5/6

Question: Why is $v^{\pi}(\mathrm{C}) = 0.5$?

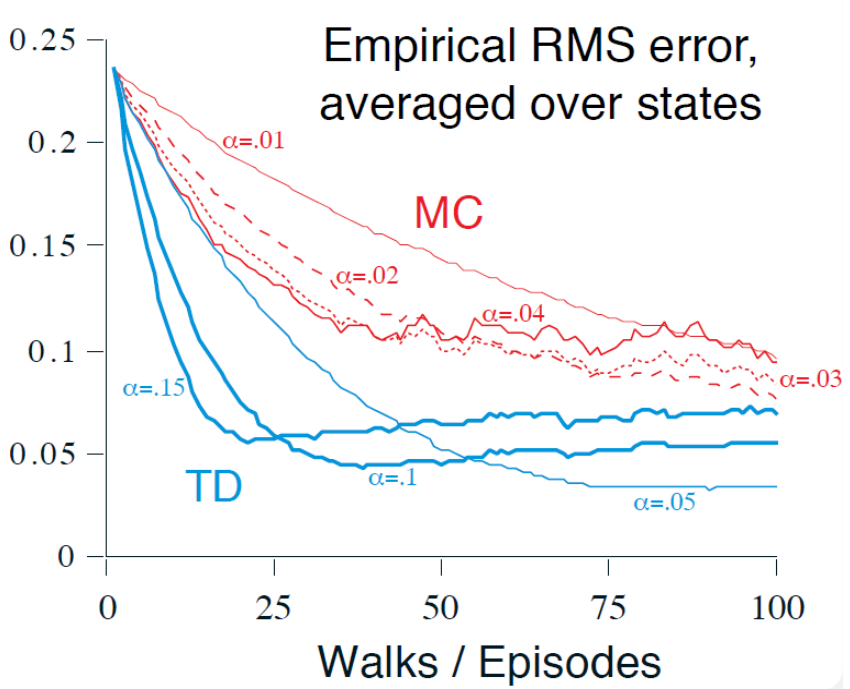
Book Example 6.2 (Sutton & Barto, 2018, p. 125).

What the random-walk curves show

Left: one TD(0) run, α=0.1, estimated values by state



Right: mean RMS error over 100 runs, several α values, by training episode



Question: After the first episode, only V(A) changed. Where did the episode end, and by how much did V(A) change?

Answer: left exit. V(A): $0.5 + 0.1(0 - 0.5) = 0.45$

Plots: book Example 6.2 (p. 125). Not Figure 6.2, which shows batch training.

Learning repeatedly from a fixed batch

We have a fixed set of complete episodes. **No new experience arrives.**

For each pass through the batch:

1. Hold V fixed while computing each visit's increment.
2. Add the increments for each state.
3. Apply those sums, then start the next pass.

$$\text{increment} = \alpha[U_t - V(S_t)]$$

MC uses the observed return. TD uses the reward and the next state's estimate.

Question: Can repeated learning from the same data lead to different answers?

Example 6.4: you are the predictor

New example: states A and B, $\gamma = 1$, zero terminal continuation.

Eight observed episodes:

Count	Episode, with rewards on transitions
1	A →(0) B →(0) terminal
6	B →(1) terminal
1	B →(0) terminal

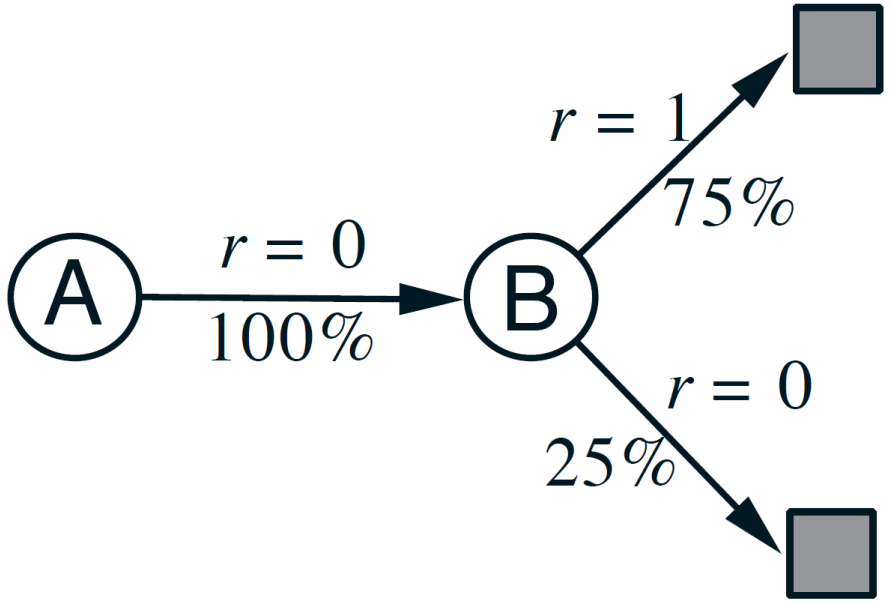
Question: What should V(A) and V(B) be after repeated batch learning?

Vote for V(A): **0**, **0.75**, or **depends on the prediction rule**.

Book Example 6.4 (Sutton & Barto, 2018, p. 127).

Two batch answers from the same data

	Batch MC	Batch TD(0)
B	$V(\mathrm{B})=6/8=0.75$	$V(\mathrm{B})=6/8=0.75$
A	Its only observed return was 0	Its observed successor was B, with reward 0
Answer at A	$V(\mathrm{A})=0$	$V(\mathrm{A})=0+V(\mathrm{B})=0.75$



Empirical model from the eight episodes

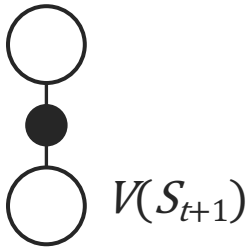
- **MC:** fit the observed returns at each state.
- **TD:** use the observed transitions and share the continuation estimate at B.

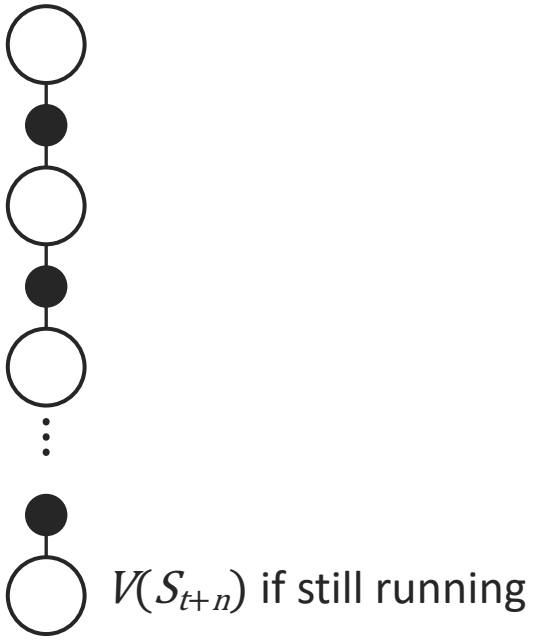
Data: A →(0) B →(0) terminal ×1; B →(1) terminal ×6; B →(0) terminal ×1.

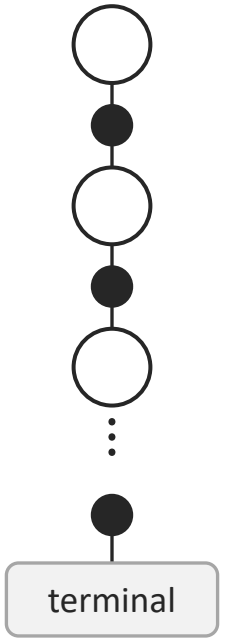
Question: When is it reasonable to reuse all the B outcomes after arriving from A?

Model picture: old course slides (book Example 6.4). Both terminal branches have continuation zero.

Between one transition and a complete return







TD(0): one reward, then V at the next state

n-step TD: n rewards, then V if the episode continues

MC: rewards through termination

Question: How much experience should we use before trusting the remaining value estimate?

Sampled backup paths after Sutton & Barto (2018), Figure 7.1. Open circles: states; solid dots: actions.

The n-step target

Before termination, replace the unobserved tail by the current estimate:

$$G_{t:t+1} = R_{t+1} + \gamma V(S_{t+1})$$

$$G_{t:t+2} = R_{t+1} + \gamma R_{t+2} + \gamma^2 V(S_{t+2})$$

For a positive integer n and a visit $t < T$:

If $t + n < T$:

$$G_{t:t+n} = \sum_{i=1}^{n} \gamma^{i-1} R_{t+i} + \gamma^n V(S_{t+n})$$

If $t + n \geq T$:

$$G_{t:t+n} = \sum_{i=1}^{T-t} \gamma^{i-1} R_{t+i} = G_t$$

Same update form when the target becomes available:

$$V(S_t) \leftarrow V(S_t) + \alpha[G_{t:t+n} - V(S_t)]$$

When can the update happen?

For the visit at time t, wait until **n further transitions or termination, whichever comes first**.

For n = 2:

Experience observed	Earliest eligible work
One transition	Wait, unless it terminated
Two transitions	Update the visit at time 0
Later transitions	Update visits as their targets become available
Episode terminates	Complete all remaining visits in their original order

Terminal completion needs **no further environment transitions**.

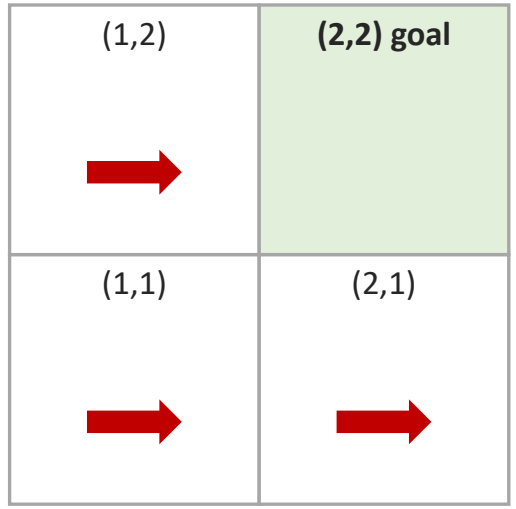
$$V(S_t) \leftarrow V(S_t) + \alpha[G_{t:t+n} - V(S_t)]$$

Your turn: targets on the grid

Always choose right. $\gamma = 0.5$, $\alpha = 0.5$. Observe:

(1,1) —(−0.04)→ (1,1) —(−0.04)→ (1,2) —(+1)→ goal, $T = 3$

Use this **frozen starting snapshot**, the table after 3.1's TD pass:



State	(1,1)	(1,2)	(2,1)	goal
V	−0.03	0.5	0	0

1. At $t = 0$, calculate the targets for n = 1, 2, 3 and 5.

2. Using n = 2, make **one isolated update** to V((1,1)).

Hold every table entry fixed while calculating part 1. Part 2 starts from the same snapshot.

Grid targets and one isolated update

Episode (1,1) →(−0.04) (1,1) →(−0.04) (1,2) →(+1) goal. Snapshot V((1,1)) = −0.03, V((1,2)) = 0.5. $\gamma = \alpha = 0.5$.

n	Target for the time-0 visit
1	−0.04 + 0.5(−0.03) = −0.055
2	−0.04 + 0.5(−0.04) + 0.5^2(0.5) = 0.065
3	−0.04 + 0.5(−0.04) + 0.5^2(1) = 0.19
5	0.19 because the episode already ended

One isolated n = 2 update:

$$V((1,1)) \leftarrow -0.03 + 0.5[0.065 - (-0.03)] = 0.0175$$

Question: For n = 5, why is there no extra discounting after the goal?

Reset, then update in time order

Reset to V((1,1)) = −0.03, V((1,2)) = 0.5. Use n = 2 and α = 0.5.

When	Visit updated	Target	New estimate
After transition 1	None	—	No change
After transition 2	t=0, (1,1)	0.065	0.0175
At termination, after transition 3	t=1, (1,1)	−0.04 + 0.5(1) = 0.46	0.23875
Finish pending work, no new transition	t=2, (1,2)	1	0.75

$$0.0175 + 0.5(0.46 - 0.0175) = 0.23875, \qquad 0.5 + 0.5(1 - 0.5) = 0.75$$

Same trajectory, delayed information, multiple visits.

A larger random walk tests the choice of n

	Earlier five-state walk	New 19-state walk
Start	C	Middle state
Moves	Left/right with probability 0.5	Same
Terminal-entry rewards	Left 0, right +1	**Left −1, right +1**
Other rewards / γ	0 / 1	0 / 1
Initial nonterminal V	0.5	**0**

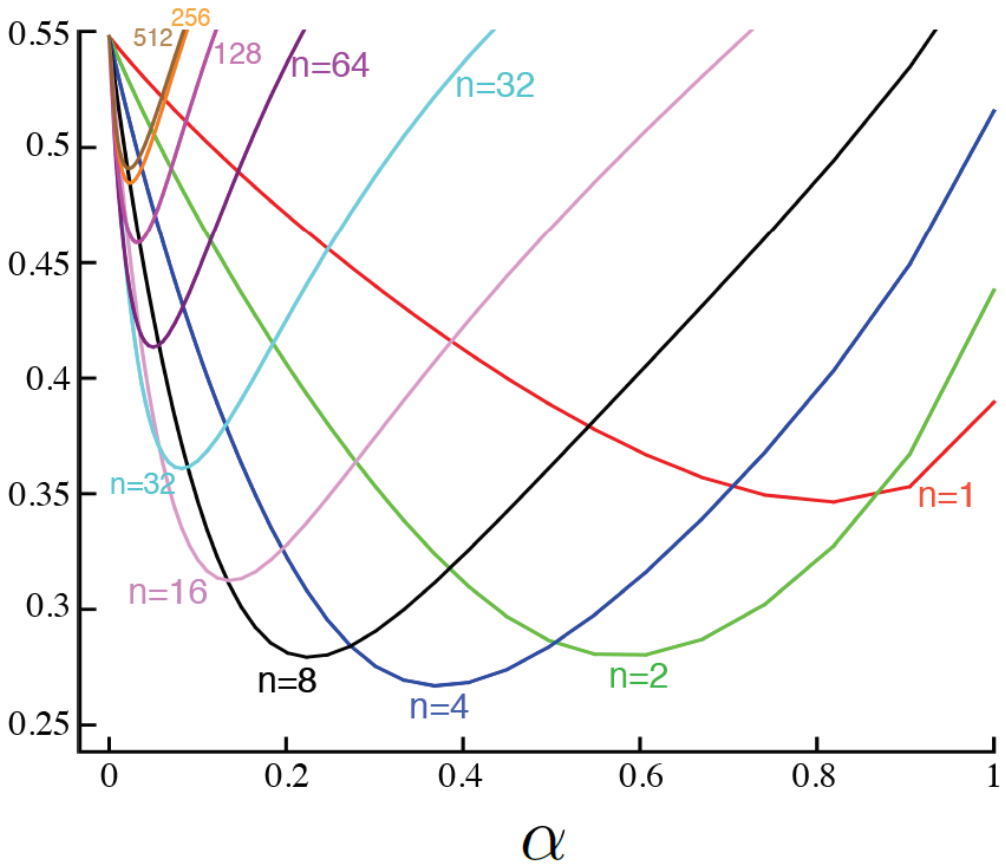
Both terminal continuation values are zero.

The plot measures RMS error over 19 states at each episode end, then averages over **the first 10 episodes and 100 runs**.

Book Example 7.1 (Sutton & Barto, 2018, p. 145).

n and step size interact

Average RMS error over 19 states and first 10 episodes



Each curve fixes n. Horizontal axis: α. Lower error is better.

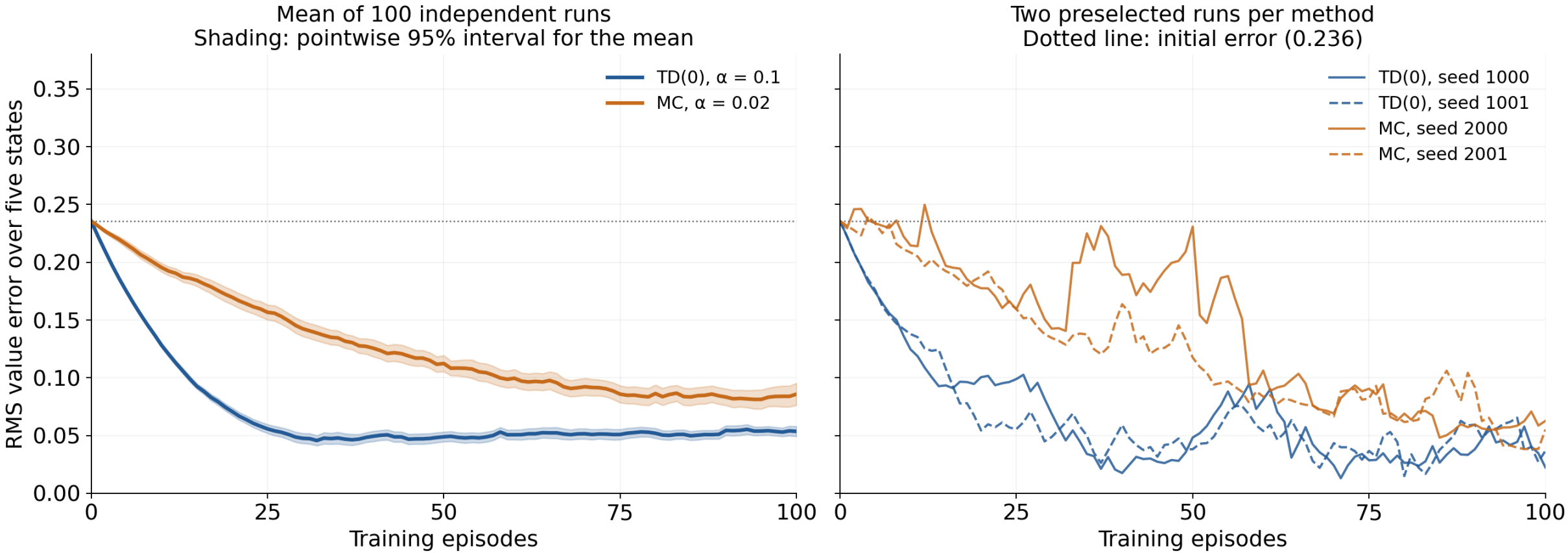
Question: Is the best n the same at every α? What can we conclude from the low points of these curves?

Intermediate n performs best **in this experiment**. Step size matters too.

The target uses the table only if the episode is still running after n steps. A longer horizon reaches a complete return more often, but can also add sampling variability and delay.

Plot: Sutton & Barto (2018), Figure 7.2 (19-state random walk).

Mean error and individual runs



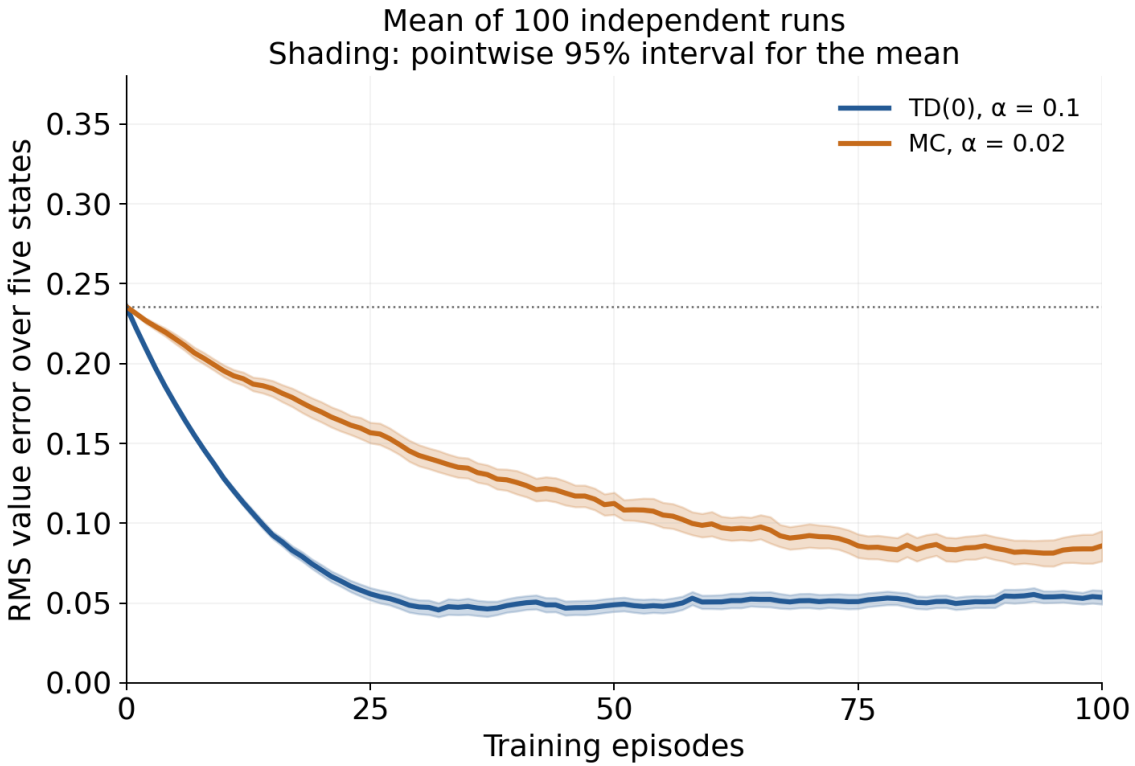
Course simulation of the five-state task. TD(0), α = 0.1; every-visit MC, α = 0.02.

Question: Do the mean-error intervals overlap at episodes 25 and 100? What does the individual-run panel tell us that the mean alone cannot?

Claim 1: “TD learns better”

“This figure proves TD is the better algorithm.”

Vote: **supported**, **unsupported**, or **needs a narrower claim**.

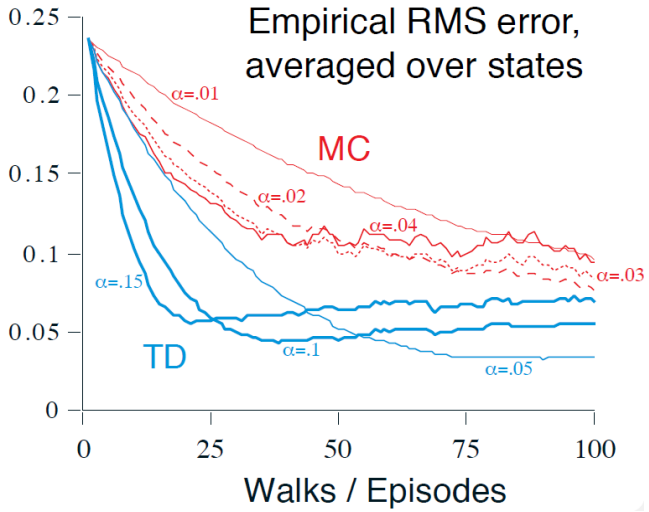


Answer: TD has lower mean prediction error at these checkpoints for **this task, these α values, initialization and episode budget**.

Question: What changes if one individual MC run has lower error than one TD run?

Claim 2: “One setting settles the comparison”

“With a wider α search, MC would win.” Vote, then identify the missing evidence.



Example 6.2 (p. 125): MC vs TD, several α

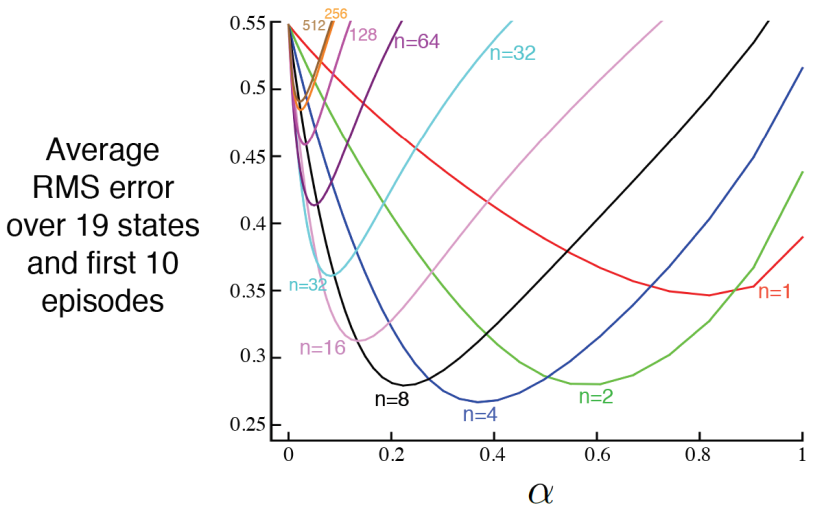


Figure 7.2: 19-state walk, n vs α

Answer: these plots do not establish that counterfactual. Compare sensitivity and give both methods **equal tuning effort**.

Question: In the 19-state experiment, how could we isolate the effect of n?

Learning: fit V using experience.

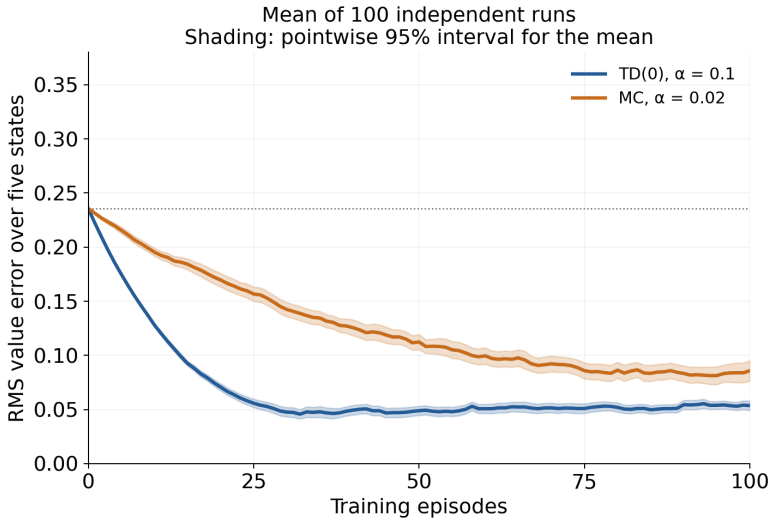
Tuning: choose α and n on separate development runs.

Final evaluation: report performance on fresh runs after those choices.

Claim 3: “Same episodes means same cost”

“Both methods used 100 episodes, so their interaction and computation costs were equal.”

Vote: **supported** or **unsupported**?



Answer: specify the budget.

Budget	What to count
Interaction	Environment transitions, not only episodes
Computation	Updates, data passes, and time under a stated implementation
Tuning	Configurations and runs used to choose them

Question: What does another pass through a fixed batch cost?

Choosing between MC, TD and n-step targets

Question	MC	TD(0)	n-step TD
Target	Complete return	One reward + V	n rewards + V if still running
Earliest update	Episode end	Next transition	n transitions or episode end
Target error	Unbiased under the fixed policy	Can inherit table bias	Horizon changes reliance on the table
Sampling variability	Often high	Often lower with a useful table	Often grows with horizon; not guaranteed
Useful when	Complete episodes are available; avoid a bootstrap target	Need immediate updates or discounted continuing prediction	Can wait for more evidence before bootstrapping
Dependence on V in the target	None	Direct	Through a surviving continuation

TD shares continuation estimates using the Markov assumption. MC's completed target avoids that bootstrap; it does not solve partial observability.

Choice needs evidence: compare runs, tuning and budgets on the task.

Next: control with sampled targets. λ mixtures: Week 8.